

Q1. Let $A = \begin{bmatrix} 0 & 0 & 0 & 0 \\ a & 0 & 0 & 0 \\ 0 & b & 0 & 0 \\ 0 & 0 & c & 0 \end{bmatrix} \in M_{4 \times 4}(\mathbb{R})$.

Under what condition on a, b and c is A diagonalizable?

Q2. If A is a 2×2 real matrix such that $A = A^T$ (Symmetric), then show that A is diagonalizable.

Q3. Let A and B be $n \times n$ matrices over $\mathbb{F}$. Show that AB and BA has same eigenvalues.

Q4. Let N be a 2×2 real matrix such that $N^2 = 0$. Show that N is similar to either 0 or $\begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$.

Q5. Let V be the space of all real-valued continuous function on $\mathbb{R}$. Define $T: V \rightarrow V$ by

$$T(f)(x) = \int_0^x f(t) dt, \quad \forall x \in \mathbb{R}.$$

Prove that the operator T (known as Volterra operator) has no eigenvalues.